

Homework Flow & Curriculum Build — Cost Verification & Provider Reconciliation

Purpose. This document lets us *confirm* the costs the estimator shows: the per-session cost of the Homework Help (Tier 2) flow (Sections 1–7), and the one-off cost to build the whole KS3 + KS4 curriculum (Section 8). It checks every provider unit price against live provider documentation, then shows exactly how the biggest homework line — **TTS audio** — is counted on the **measured average** narration length, how image generation is costed at **one diagram per session** on the new **Gemini 3.1 Flash Image Light** model, and explains the o3 reasoning-token risk.

Bottom line up front: the unit prices are *correct and current*. Audio is billed on the **MEASURED AVERAGE** narration length (Section 2), route-specific — measured across real homework sessions rather than the old conservative 24,000-char maximum. Homework prices carry a **+15% buffer** and Inworld is priced on the **Developer plan (\$17 / 1M characters)**. Image generation is a single diagram per session on the new **Gemini 3.1 Flash Image Light (Nano-Banana-Light)** model at **\$0.05/image** (Section 3), down from \$0.10. The figure to keep validating on a real invoice is o3's **hidden reasoning tokens** (Section 4.3).

1. Provider unit prices — confirmed against documentation (July 2026)

Service	Model	Published list price	In estimator	Confirmed
Reasoning LLM	o3	\$2.00 / 1M input · \$8.00 / 1M output	\$2 / \$8	✓ matches OpenAI
Prose / vision LLM	gpt-4.1	\$2.00 / 1M input · \$8.00 / 1M output	\$2 / \$8	✓ matches OpenAI
Diagram image	Gemini 3.1 Flash Image Light (Nano-Banana-Light)	\$0.05 / image	\$0.05	✓ matches
Text-to-speech	Inworld TTS-2	\$17.00 / 1M characters (Developer plan)	\$17	✓ matches Inworld
Safety	omni-moderation	Free	\$0	✓

The +15% safety buffer (homework). Every homework price above is multiplied by **1.15** inside the estimator before any figure is shown, as an internal margin for retries, prompt drift and future price moves. So the effective rates the estimator charges against are: o3 / gpt-4.1 **\$2.30 / \$9.20** per 1M, Gemini **\$0.0575** per image, Inworld **\$19.55** per 1M characters. Throughout, we show **both** columns — "*list*" (the provider's published price, for reconciliation) and "+15%" (what the estimator displays). The one-off **curriculum build** (Section 8) uses a *different* buffer regime — see there.

2. Text-to-speech (Inworld TTS-2) — billed on the MEASURED AVERAGE narration length

Inworld bills **per character** sent to synthesis. Earlier this document costed audio on a conservative **24,000-char maximum**. We have since **measured** the characters the homework flow actually sends to TTS across real sessions (`homework_tts_char_count.py` , 6 sessions), and the estimator now costs audio on that **measured average** — and, crucially, **route-specific**: prose *explanation* sessions narrate 2–4× longer than *calculation* sessions, so the blended figure follows the calculation / explanation mix.

Route (what the question needs)	Measured avg TTS chars / session	% of the 24,000 ceiling	n
Calculation → o3 (Maths / Physics / Chemistry sums)	4,758	20%	3
Explanation → gpt-4.1 (English, Biology prose)	12,611	53%	3
Blended average (50/50 default)	8,685	36%	6

Cost of the blended 8,685-char average: list $8,685 \div 1,000,000 \times \$17 = \$0.1476 \cdot +15\% = \0.1698 . Per route: calculation $4,758 \times \$17/1M = \0.0809 list / **\$0.0930** (+15%); explanation $12,611 \times \$17/1M = \0.2144 list / **\$0.2465** (+15%).

The 24,000-char maximum remains a safe worst-case bound, but the measured average is **~2.8× lower**, so it is the realistic figure for a blended forecast. Inworld also gets cheaper with commitment, so a real audio invoice can only come in lower than the Developer-plan rate:

Inworld TTS-2 tier	\$ / 1M characters	Cost of the 8,685-char avg session
On-demand	\$25.00	\$0.217
Developer (what we estimate on)	\$17.00	\$0.148
Higher commit	\$12.50	\$0.109
Enterprise	\$10.00 (as low as \$5)	\$0.087 (↓ \$0.043)

3. Image generation (Gemini 3.1 Flash Image Light) — one diagram per session

Homework Help generates **exactly one diagram image per session**, regardless of walkthrough depth or how the question arrived. This is homework-only behaviour: a single supporting diagram is produced for the session and reused across the walkthrough steps rather than re-generated at each phase.

```
images / session = 1    (fixed – one diagram per session)
```

Because it no longer scales with walkthrough depth, this line is flat and small. The homework flow has moved to the new **Gemini 3.1 Flash Image Light (Nano-Banana-Light)** model, so the estimator now prices the image at **\$0.05** — down from **\$0.10** on the previous model:

Images / session	Cost (list @ \$0.05)	Cost (+15%)
1 (fixed)	\$0.05	\$0.0575

This is a minor provider line in a session. At one image, generation is **\$0.05 list / \$0.0575 buffered**. If in future the flow needs more than one diagram per session, raise the image assumption in the estimator and this line scales linearly.

4. o3 reasoning model — tokens and cost

o3 is only used for **calculation / derivation** questions (Maths, Physics, Chemistry, computational Science). Prose/explanation questions run on gpt-4.1. The build pipeline is five LLM calls (steps A · parallel B · demo C · 3A explanation · answer key D); A→B→C can regenerate on a self-check failure, so they carry a retry factor (default **1.15×**).

4.1 Per-call token assumptions (o3 route)

Output tokens below are set to **include** o3's hidden reasoning tokens — see Section 4.3.

Call	Input tok	o3 output tok	Runs / session	o3 cost (list)
A · Solution steps	700	3,000	1.15	\$0.0292
B · Parallel question	800	2,500	1.15	\$0.0248
C · Parallel demo (worked)	900	3,000	1.15	\$0.0297
3A · Explanation	600	2,200	1.00	\$0.0188
D · Private answer key	900	3,000	1.00	\$0.0258
Validate attempt	700	1,500	3.00	\$0.0402
Follow-up Q&A	800	1,500	1.00	\$0.0136
o3 subtotal	~7,160	~20,975		\$0.1821

Plus the fixed gpt-4.1 intake/classify calls that run in **every** session (image extract + analyze + match): **\$0.0118**.

4.2 Per-session o3 cost (the math)

o3 input : 7,160 tok ÷ 1,000,000 × \$2 = \$0.0143

o3 output: 20,975 tok ÷ 1,000,000 × \$8 = \$0.1678

o3 total (list) = \$0.1821 → +15% = \$0.2094

4.3 Why o3 can look cheaper than the docs — hidden reasoning tokens

The unit price is right (\$2/\$8); the risk is the token count, not the rate. o3 bills *invisible* reasoning tokens at the **output** rate. OpenAI's documentation notes a single o3 answer can burn **5,000–20,000 internal reasoning tokens**. Our model already assumes ~21,000 o3 output tokens for a whole calculation session (near the high end), but if o3 reasons *harder*, cost scales. This table shows a full **calculation** session (o3 LLM + the fixed gpt-4.1 intake + the measured-average calc audio \$0.081 + the one-image \$0.05 line) as reasoning load rises:

o3 reasoning load	o3 output tok / session	Calc session (list)	Calc session (+15%)
As modelled	~21,000	\$0.325	\$0.374
1.5× harder	~31,500	\$0.409	\$0.470
2× harder	~42,000	\$0.493	\$0.567
3× harder	~63,000	\$0.661	\$0.760

The o3 output-token counts are **editable** in the estimator's *Advanced assumptions* — tune them to a sample of real billed sessions and the estimate tracks reality.

5. Full homework-session confirmation table

Per-session cost by service, with **audio at the measured average** (route-specific) and **one real-time image**. **Calculation** sessions use o3; **explanation** sessions use gpt-4.1; the **blended** row is the estimator default (50/50, image intake, audio on).

Session type	o3	gpt-4.1	Gemini image (×1)	Inworld audio (avg)	Total (list)	Total (+15%)
Calculation (o3)	\$0.182	\$0.012	\$0.050	\$0.081	\$0.325	\$0.374
Explanation (gpt-4.1)	—	\$0.070	\$0.050	\$0.214	\$0.334	\$0.384
Blended (default)	\$0.091	\$0.041	\$0.050	\$0.148	\$0.330	\$0.379

Every number is reproducible from Sections 2–4 with the published unit prices — that is the confirmation: **provider price × quantity = the estimator's figure**. Audio is now a measured line (~35–55% of a session depending on route), not a worst-case ceiling; the single image line is minor.

6. Scaling to volume (monthly / yearly)

Using the **blended buffered** cost of **\$0.379** per session, scaled by each preset's monthly session count (the *range* uses the calculation-only and explanation-only bounds, \$0.374 → \$0.384):

Preset	Students × questions	Sessions / month	Monthly (blended)	Yearly (blended)	Monthly range
Reset to defaults	200 × 10	2,000	\$758	\$9,096	\$748 – \$768
Light usage	300 × 20	6,000	\$2,274	\$27,288	\$2,244 – \$2,304
Heavy usage	600 × 40	24,000	\$9,096	\$109,152	\$8,976 – \$9,216

Monthly figures use the blended-default per-session cost. The two levers that move these most: the explanation / calculation mix (prose sessions narrate 2–4× more audio) and the o3 reasoning-token count (Section 4.3). The Inworld commitment tier lowers audio further (\$17 Developer → \$12.50 higher commit → \$10 enterprise).

7. Reconciliation summary (homework)

- Unit prices are correct and current** (verified July 2026) — o3 & gpt-4.1 at \$2/\$8 per 1M tokens, Gemini 3.1 Flash Image Light at \$0.05/image, Inworld TTS-2 at **\$17 per 1M characters (Developer plan)**.
- Audio is now a measured-average line, route-specific** — calculation ~4,758 chars, explanation ~12,611 chars, blended ~8,685 chars ≈ \$0.148 list / \$0.170 buffered — measured across 6 real sessions, replacing the old 24,000-char maximum (which remains a safe worst-case bound, ~2.8× higher).
- Image generation is a single diagram per session** — one **Gemini 3.1 Flash Image Light (Nano-Banana-Light)** image at **\$0.05 list / \$0.0575 buffered**, down from \$0.10, a minor line that no longer scales with walkthrough depth.
- Homework carries a +15% buffer**, so the estimate stays deliberately conservative.
- Validate two things on real usage:** (a) the **average narrated characters** per session vs. the measured 4,758 / 12,611 / 8,685 figures — the dominant line; and (b) o3's reasoning-token count (Section 4.3).

Recommended checks before committing to a budget

- Confirm the **average narrated characters** per session (per route) against a larger sample than 6.
- Pull the **actual o3 output-token count** from a dozen real calculation sessions vs. ~21,000/session.

- Decide the Inworld **commitment tier** (Developer \$17 → higher commit \$12.50 → enterprise \$10).

8. Full curriculum build — one-off content generation (separate from homework)

This is **not** a per-homework-session cost. It is the **one-off** cost to generate every lesson across the KS3 + KS4 curriculum, costed on **gpt-4.1** (the most expensive content model). Like homework, this build has now moved to the cheaper new image model for its ~30 whiteboard images per lesson.

8.1 Basis and buffer regime

The source is the *General Cost Estimation Document (Updated).pdf*, whose per-lesson **gpt-4.1** cost is **\$3.26 – \$4.06** — and that figure already reflects the new image model (30 images ~\$1.05, down from ~\$1.89 on the old model). The estimator then applies **two** things (a different regime from the +15% homework buffer):

- a **×1.5 scope/contingency multiplier** over the source per-lesson figure (headroom for the 3 mastery levels, scope-audit regenerations, and drift), and
- a **+19.5% low→high buffer**, held constant, as the displayed range.

per-lesson low	= \$3.26 (Updated PDF, gpt-4.1)	× 1.5	= \$4.89
per-lesson high	= \$4.89	× 1.195 (maintained 19.5% buffer)	= \$5.84
images/lesson	= ~30 on Gemini 3.1 Flash Image,	~\$1.05 raw → ~\$1.58 (×1.5)	

Image-model change, isolated. The only input that moved from the previous build is the image cost: 30 images **\$1.89 → \$1.05** (raw), i.e. **\$2.84 → \$1.58/lesson** on the ×1.5 basis. That drops the per-lesson low from **\$6.15 → \$4.89**; the high is then held at low × 1.195. Because images are a *fixed* cost, holding the spread at 19.5% pulls the high slightly tighter than a raw subtraction would.

8.2 Per key stage

Key Stage 3

Subject	Units	Lessons	Cost
Mathematics	6	57	\$278.7 – \$333.1
English	8	27	\$132.0 – \$157.8
Physics	7	43	\$210.3 – \$251.3
Chemistry	8	32	\$156.5 – \$187.0
Biology	6	38	\$185.8 – \$222.1
Total	35	197	\$963.3 – \$1,151.3

Key Stage 4

Subject	Units	Lessons	Cost
Mathematics	6	74	\$361.9 – \$432.5
English	11	38	\$185.8 – \$222.1
Physics	7	34	\$166.3 – \$198.7
Chemistry	10	49	\$239.6 – \$286.4
Biology	—	—	—
Total	34	195	\$953.5 – \$1,139.6

8.3 Whole curriculum

Whole curriculum (KS3 + KS4)	Units	Lessons	Cost
Combined	69	392	\$1,916.9 – \$2,290.9

Down from \$2,410.8 – \$2,881.2 on the previous image model. Biology KS4 was not in the source curriculum count, so it is left blank rather than estimated. The combined total is the sum of the two Key-Stage totals.

Sources

- OpenAI o3 API pricing (\$2 / \$8 per 1M tokens; hidden reasoning tokens billed at output rate): <https://developers.openai.com/api/docs/pricing> · <https://pricepertoken.com/pricing-page/model/openai-o3>

- Inworld TTS-2 pricing (Developer plan \$17 / 1M characters; on-demand \$25, tiered to \$10 enterprise): <https://inworld.ai/pricing> · <https://www.buildmvpfast.com/api-costs/ai-voice>
- Measured TTS characters: `homework_tts_summary.pdf` / `homework_tts_char_count.py` (6 sessions, 2026-07-10).
- Curriculum build figures: `General Cost Estimation Document (Updated).pdf` (per-lesson gpt-4.1 with the new image model), reconciled against `General Cost Estimation Document.pdf` (previous image model).

Figures are grounded in the estimator's editable token / character / image assumptions; tune those to your real provider contracts and billed usage.